

Test Report No. 7191279319-MEC22/01A-ES (5543024)
dated 24 May 2022



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SUBJECT:

Testing of sealant submitted by Bostik Findley (M) Sdn Bhd.

TESTED FOR:

Bostik Findley (M) Sdn Bhd
Lot 112 & 113, Kawasan Perindustrian Senawang
70450 Seremban, Negeri Sembilan
Malaysia

Attn: Mr. Herman Chen / Mr. Wang SM

SAMPLE DESCRIPTION:

The following items were received on 23 Dec 2021 as shown:

Sample/Material	Size	Quantity
'Bostik S548 Glass 'N' Seal Weatherseal Neutral Silicone For Weather Sealing & Window Sealing' (refer to Photo 1)	260 ml/cartridge	10 cartridges



Photo 1: 'Bostik S548 Glass 'N' Seal Weatherseal Neutral Silicone For Weather Sealing & Window Sealing'



LA-2007-0380-A
LA-2007-0381-F
LA-2007-0382-B
LA-2007-0383-G
LA-2007-0384-G
LA-2007-0385-E
LA-2007-0386-C
LA-2010-0464-D
LA-2018-0702-B
LA-2018-0703-G
LA-2020-0747-L

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TUV®

TEST METHODS:

HDB Specifications Sealant - Semi-Precast Construction

Staining And Colour Change, UV Exposure

1. Adopted ASTM C510 : 2016 Standard Test Method For Staining And Colour Change Of Single Or Multi-Component Joint Sealants

Test equipment	:	QUV Weatherometer
Lamp designation	:	Fluorescent UVA 340 mm
Test cycle	:	8 hours UV exposure at 60±3°C and 4 hours condensation at 50±3°C, irradiance 0.89 W/m ² .nm (ASTM G154)
Exposure duration	:	100 hours
No. of determinations	:	4 samples: 2 samples with sealant and 2 samples without sealant (For UV Exposure) 2 control samples: 1 sample with sealant and 1 sample without sealant (Standard Conditions)

Staining And Colour Change, Standard Conditions In Distilled Water

Test apparatus	:	Container with distilled water
Test condition	:	Distilled water immersion for 1 minute, once a day, (5 days per week)
Test duration	:	14 days
No. of determinations	:	2 samples: 1 sample with sealant and 1 sample without sealant (For distilled water immersion) 2 control samples: 1 sample with sealant and 1 sample without sealant (Standard Conditions)

Extrudability

2. Adopted ASTM C1183/C1183M : 2013 (2018) Standard Test Method For Extrusion Rate Of Elastomeric Sealants

Test pressure	:	40 psi
No. of determination	:	1

Flow Properties

3. ASTM C639 : 2015 Standard Test Method For Rheological (Flow) Properties Of Elastomeric Sealants

Method	:	Test method for 'Type II' sealant
Test conditions	:	a) 4.4°C in environmental chamber for 4 hours b) 50°C in oven for 4 hours
No. of determinations	:	2 for vertical and horizontal displacements





TEST METHODS: (cont'd)

Hardness

4. ASTM C661 : 2015 Standard Test Method For Indentation Hardness Of Elastomeric-Type Sealants By Means Of A Durometer

Test Conditions:

- a) 23°C and 50% relative humidity for 7 days
 - b) 38°C and 95% relative humidity for 7 days
 - c) 23°C and 50% relative humidity for 7 days
- No. of determinations : 2, 3 points per test piece

Tack-Free Time

5. ASTM C679 : 2015 Standard Test Method For Tack-Free Time Of Elastomeric Sealants

No. of determinations : 2

Cyclic Adhesion & Cohesion

6. Adopted ASTM C719 : 2014 (2019) Standard Test Method For Adhesion And Cohesion Of Elastomeric Joint Sealants Under Cyclic Movement (Hockman Cycle)

Test Conditions:

- a) 23°C and 50% relative humidity for 7 days
 - b) 38°C and 95% relative humidity for 7 days
 - c) 23°C and 50% relative humidity for 7 days
 - d) Immersion in distilled water at 23°C for 7 days
 - e) Drying in oven at 70°C for 7 days
- Substrate : Aluminium and glass
Test temperature : Room temperature
Class : 25
No. of cycles : 10
Crosshead speed : 3.2 mm/hr
No. of determinations : 3 per substrate

Effects Of Heat Ageing

7. ASTM C1246 : 2017 Standard Test Method For Effects Of Heat Ageing On Weight Loss, Cracking, And Chalking Of Elastomeric Sealants After Cure

Test Conditions:

- a) 23°C and 50% relative humidity for 28 days
 - b) 70°C for 21 days
- No. of determinations : 3, 1 as control

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TEST METHODS: (cont'd)

Effects Of Accelerated Weathering

8. Adopted ASTM C793 : 2005 (2017) Standard Test Method For Effects Of Accelerated Weathering On Elastomeric Joint Sealants

Test Conditions:

23°C and 50% relative humidity for 21 days

Test equipment : QUV Weatherometer
Test cycle : 8 hours UV exposure at 60±3°C and 4 hours condensation at 50±3°C, irradiance 0.89 W/m².nm (ASTM G154)
Lamp designation : Fluorescent UVA 340 nm

Exposure duration : 250 hours
No. of determinations : 3 (1 as control)
Bend test
Test equipment : Environmental chamber
Apparatus : Steel mandrel
Test condition : -26°C for 24 hours
No. of determinations : 3

Adhesion-In-Peel

9. ASTM C794 : 2018 Standard Test Method For Adhesion-In-Peel Of Elastomeric Joint Sealants

Test Conditions:

23°C and 50% relative humidity for 21 days

Substrate : Aluminium and glass
Crosshead speed : 50 mm/min
No. of determinations : 4 per substrate

CONDITIONING:

Unless otherwise specified, all test specimens were tested at 23 ± 2°C and 50 ± 5% relative humidity.
Standard Conditions parameters: 23 ± 2°C and 50 ± 5% relative humidity.

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TEST RESULTS:

Test	'Bostik S548 Glass 'N' Seal Weatherseal Neutral Silicone For Weather Sealing & Window Sealing'	HDB Specifications Sealant - Semi-Precast Construction
1. Staining And Colour Change	No staining No colour change	No visible staining on white cement mortar base
2. Extrudability	113.4 ml/min	> 10 ml/min
3. Rheological (Flow) Properties	Vertical displacement: 0 mm sag Horizontal displacement: No deformation	Vertical displacement: < 4.8 mm Horizontal displacement: no deformation
4. Indentation Hardness	test piece 1, average : 47.6 test piece 2, average : 45.6 average of 2 test pieces : 46.6	25 to 50 (traffic) 15 to 50 (non-traffic)
5. Tack-Free Time	No transfer of test specimens to the polyethylene film after 50 mins	No transfer of sealant to PE film
6. Adhesion & Cohesion Under Cyclic Movement, Class 25 a. Aluminium b. Glass	No loss in bond No loss in bond	Total loss in bond and adhesion < 9 cm ²
7. Effects Of Heat Ageing On Weight Loss, Cracking And Chalking, average	0.3% No cracking and chalking	Loss in weight < 7% No cracking and chalking
8. Effects Of Accelerated Weathering	No cracks after UV exposure and bend test	No cracks
9. Adhesion-In-Peel, average a. Aluminium b. Glass	104.9 N 115.2 N cohesive failure within the sealant and no adhesive bond loss between sealant and substrate for each test piece	Peel strength >22.2 N Bond loss <25%

REMARKS:

- The test conditions for staining and colour change tests and effects of accelerated weathering test were adopted from ASTM G154 : 2016 Standard Practice For Operating Fluorescent Light Apparatus For UV Exposure Of Non-Metallic Materials.
- The tests and HDB specifications are specified by the client.
- The substrates did not require primer before application of the sealant as specified by the client.
- The class and types of substrate are specified by the client for ASTM C719 joint movement and ASTM C794 peel strength tests.
- One cartridge was sent to other section for material identification/verification FTIR test.
- Note in HDB specifications:
 - The sample shall be tested for new and renewal listing.
 - The sample for testing shall be selected by officers from HDB.


Lem Chee Meng
Testing Officer


Eddie Suwand
Senior Associate Engineer
Real Estate & Infrastructure
Mechanical Centre

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Effective 26 January 2021

